FORMAL ASSESSMENT
NATURAL SCIENCES
Grade 7 Term: 4 Task 1
Controlled Test Memorandum (02)

<u>Total:</u> 80

Topics	Question Number	Cognitive Levels			Total Mark
		Low (40%)	*Middle (45%)*	*High (15%)*	
Section A: Concepts in Term 3 & 4					
Multiple choice	1.1	10			**10**
Match Column A with Column B	1.2	5			**5**
One word answer	1.3	5			**5**
	Subtotal	20			**20**
				Total of Section A	
Section B: Energy and Change Term 3					
Potential and kinetic energy	2.1	1			**10**
	2.2			2	
	2.3		3		
	2.4	3	1		
Heat transfer	3.1		6		**10**
	3.2			2	
	3.3			2	
Insulation and energy saving	4.1		1		**10**
	4.2		1	1	
	4.3		5		
	4.4		2		
Energy transfer to surroundings	5.1	2			**6**
	5.2		1		
	5.3		1		
	5.4			2	
	Subtotal	6	21	9	**36**
				Total of Section B	
Section C: Planet Earth and Beyond Term 4					
Relationship of the Sun to the Earth	6.1	2	4		**15**
	6.2		2	3	
	6.3		2		
	6.4		2		
Relationship of the Moon to the Earth	7.1	1			**9**
	7.2	1			
	7.3	1	1		
	7.4	1			
	7.5		4		
	Subtotal	6	15	3	**24**
				Total of Section C	
Total		**32**	**36**	**12**	**80**

Section A:

Question 1:

1.1. **Choose the correct answer by only circling the letter e.g. 1.7 A.**

1.1.1. Which of the following is an example of wasted energy in a light bulb? (1)

A. Light
B. Heat✓
C. Electricity
D. Sound

1.1.2. Which of the following is a NON-RENEWABLE energy source? (1)

A. Solar power
B. Wind power
C. Coal ✓
D. Hydropower

1.1.3. How does burning fossil fuels, like coal and oil, contribute to environmental problems? (1)

A. They release greenhouse gases that trap heat in the atmosphere. ✓
B. They create a limited supply of energy for the future.
C. They are not very efficient at producing energy.
D. They are difficult to transport over long distances.

1.1.4. A book is sitting on a shelf. What type of energy does the book have? (1)

A. Kinetic energy
B. Thermal energy
C. Light energy
D. Potential energy✓

1.1.5. When a stretched rubber band is released, what happens to its potential energy? (1)

A. It is converted to kinetic energy✓
B. It increases
C. It stays the same
D. It disappears

1.1.6. A battery powers a toy car. The battery loses energy as the car moves. What happens to the total amount of energy in the system (battery and car)? (1)

A. The total energy increases.
B. The total energy decreases.
C. The total energy stays the same, but it changes form. ✓
D. The energy disappears

1.1.7. The Earth revolves around the Sun in a path called its: (1)

A. Rotation
B. Tilt
C. Rotation
D. Orbit✓

1.1.8. Why do we experience different seasons on Earth? (1)

A. The Sun gets hotter and colder throughout the year.
B. The Earth's distance to the Sun changes throughout the year.
C. The Earth's axis is tilted as it revolves around the Sun.✓
D. The Moon blocks the Sun's rays for part of the year.

1.1.9. What is the process called where plants capture the sun's energy? (1)

A. Combustion
B. Photosynthesis✓
C. Respiration
D. Transpiration

1.1.10. Which of the following statements is TRUE about gravity? (1)

A. It pulls objects with a greater mass towards objects with a lesser mass.✓
B. It only affects very large objects like planets and stars.
C. It gets stronger the further apart two objects are.
D. It only exists on Earth

1.2. Match the term in column A with the definition in Column B. Write the answer below. (5)

Column A	Column B
1.2.1. Conduction	A. A current in a fluid that results from convection.
1.2.2. Convection current	B. An expert in the field of geology, the study of what the earth is made of and how it was formed.
1.2.3. Solar power	C. A person who is trained to travel in a spacecraft.
1.2.4. Geologist	D. Power obtained by harnessing the energy of the sun's rays.
1.2.5. Astronaut	E. The process of which heat, or electricity passes through a material.

1.2.1. E✓	1.2.2. A✓	1.2.3. D✓	1.2.4. B✓	1.2.5. C✓

1.3. Give one word for the following definition: (5)

1.3.1. Power obtained by harnessing the energy produced by waves at sea.	wave power✓
1.3.2. Potential energy that is stored when a body is deformed.	elastic potential energy✓
1.3.3. Energy produced by heat.	thermal energy✓
1.3.4. An imaginary line around which a body rotates.	axis✓
1.3.5. A series of organisms each dependent on the next as a source of food.	food chain✓

Section B:

Question 2:

Answer the following questions.

2.1. Define kinetic energy. (1)

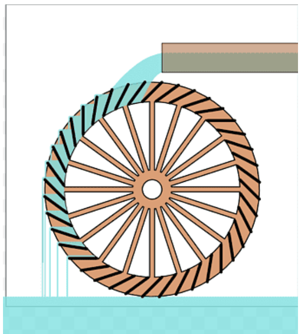
Kinetic energy is the energy an object possesses due to its motion.✓

2.2. Look at the picture of the two vehicles.

2.2.1. Why does the truck at the top of the hill have more potential energy than a car speeding down the hill? (2)

The truck has more potential energy because the potential energy✓ is directly related to the height of an object.✓

2.3. Look at the picture of a water wheel.



[PNGWING, *Waterwheel,* 25 June 2024, < https://www.pngwing.com/en/search?q=water+Wheel>]

2.3.1. Fill in the flow diagram. (3)

Possible answer:✓✓✓

Input

- The water at a height, has gravitational potential energy.

Process

- Gravity lets the water fall onto the fins of the water wheel. Gravitational potential energy is changed to kinetic energy.

Output

- The falling water causes the wheel to turn.

2.4. Read the case study and answer the questions that follow.

Water Reservoirs

Water reservoirs are built to provide water for towns. They are built uphill from the houses in a town. In this way, the stored water has enough potential energy to flow when a tap in the town is opened. The height of the reservoir determines how strongly the water will flow from the taps.

When the water reservoir is not much higher than the houses, the water only trickles out of the taps. Water in reservoirs that are positioned lower than the houses, or water from boreholes and wells, must be pumped up to the houses. This is because the water below the level of the houses does not have enough energy to reach the houses.

[Laerskool Van Dyk Primary, 27 June 2024, <https://drive.google.com/file/d/1MroBwwAr_uByiIKWfBrCgjsXh-iqa3py/view>]

2.4.1. What are the three types of potential energy? (3)

elastic potential energy✓
gravitational potential energy✓
chemical potential energy✓

2.4.2. What type of potential energy is a water reservoir? (1)

gravitational potential energy✓

[10]

<u>Question 3</u>:
Answer the questions below.

3.1. Explain the principles of convection and heat energy in the following situations. (6)

	The water at the bottom of the pot becomes hotter,✓ more energetic and less dense than the rest of the water.✓ Currents of the hot water rise, pushing aside the colder water at the top of the pot.✓

	Even though your body generates heat, still air conducts it away slowly.✓ Wind speeds up this transfer by constantly✓ replacing warm air around you with cooler air, making you feel colder.✓

3.2. You're building a doghouse in a hot climate and want to keep it cool in the summer, how would you choose a roof colour to minimize heat gain inside the doghouse. Give an explanation. (2)

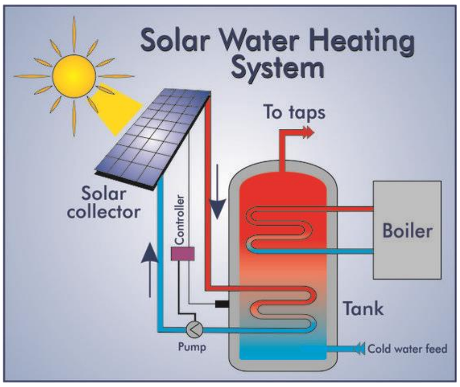
I would choose a light-coloured roof✓ because lighter colours reflect the heat most, preventing the object from heating up as much.✓

3.3. There are three spoons (plastic, metal and wooden spoon) in the bowl. Which spoon would be the best conductor of heat? Give a detailed explanation. (2)

Metal is a good conductor of heat✓ and plastic and wood are not good conductors.✓

[10]

<u>Question 4</u>:



[Pinterest, 27 June 2024, <https://za.pinterest.com/pin/415668240579238791/>]

4.1. Why do you think the pipes and tank in a solar water heating system are insulated. (1)

The pipes and tank are insulated to keep the water hot for longer.✓

4.2. You have bought a second-hand solar water system at a very good price. After a few days, you notice that the system is heating up the water but not keeping it hot. What could the problem be? Give a detailed explanation. (2)

Possible answer

Poor insulation✓ could be the problem, because hot water loses heat as it travels through the pipes to your taps.✓

4.3. Discuss how a solar water heating system would work, after analysing the above image. (5)

A solar heating system heats tap water from the solar collector / the Sun. ✓ Most homes have electricity, but a solar water heating system uses radiation. ✓At the core of a solar water heating system, is a solar collector and a storage tank.✓ Cold water will be pumped into the tank and go through a boiler. ✓ The pump controller will heat it up and you will get hot water.✓

4.4. Based on the solar water heating system, explain how this system can save energy compared to traditional water heating methods using electricity. (2)

Possible answer

The solar system uses renewable solar energy to heat water, ✓reducing the need to use electricity.✓

[10]

<u>Question 5</u>:

Answer the following questions.

5.1. Underline two examples of useful energy. (2)

a. <u>Light bulb illuminating a room</u>✓
b. Smoke rising from a chimney
c. <u>Food giving you energy to run</u>✓
d. Leaving a light on in an empty room

5.2. Identify one way you can conserve energy at home. (1)

Possible answer

Turning off the lights in empty rooms or switching off electronics.

5.3. Does electricity get wasted when an appliance is on standby mode? Give a reason for your answer. (1)

Possible answer:

Yes, electricity is wasted when appliances are on standby mode, because there is a constant draw on the electricity grid.✓

5.4. You notice your phone gets warm while charging. Is this useful or wasted energy or both? Give an explanation. (2)

It is both, the electricity is converted to heat and that is wasted✓ but also to stored energy in the phone battery that is useful.✓

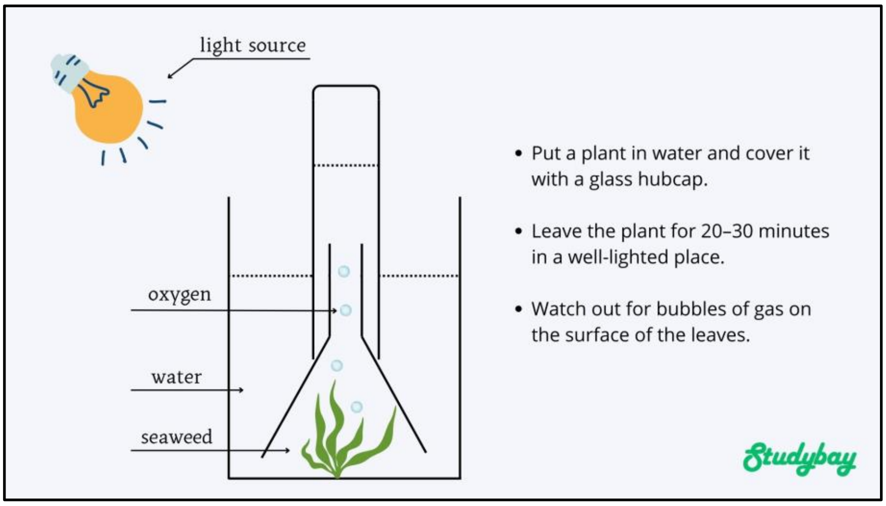
[6]

Section C:

Question 6:

Answer the questions below.

6.1. Read the description of the experiment and the sketch.



[Studybay, 27 June 2024, < https://studybay.com/blog/photosynthesis-lab-report/>]

6.1.1. What is the process by which plants produce food? (1)

photosynthesis✓

6.1.2. What gas does the plant take in during this process? (1)

carbon dioxide✓

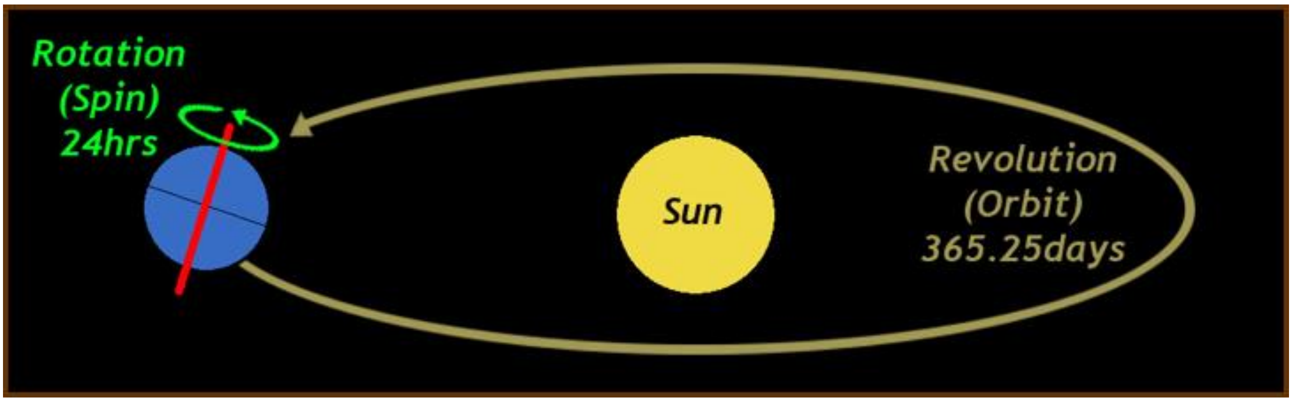
6.1.3. In the image, what is the purpose of placing the plant in a container of water and covering it? (1)

To see the oxygen gas bubbles. ✓

6.1.4. Based on your knowledge of photosynthesis, what are the raw materials that a plant needs to carry out this process? (3)

water, ✓ carbon dioxide✓ and sunlight. ✓

6.2. Look at the diagram and answer the questions below.



[Weebly, 27 June 2024, < https://planetearth1p.weebly.com/earths-revolution.html>]

6.2.1. Earth's orbit around the Sun is not perfectly circular, but slightly elliptical. How might this slight difference in Earth's distance to the Sun throughout the year influence seasonal temperature.?. (3)

The elliptical orbit has a minor effect on seasonal temperatures.✓ When Earth is closer to the Sun (perihelion), it receives slightly more sunlight, which could contribute to a warmer season in the hemisphere pointed more towards the Sun.✓ However, the primary reason for seasonal variations in temperature is the tilt of Earth's axis, not the slight change in distance due to the elliptical orbit.✓

6.2.2. Why do the seasons differ in the different hemispheres? (2)

This is because the tilt of the Earth's axis as it orbits✓ the Sun causes different hemispheres to receive more direct sunlight at different times of the year.✓

6.3. Where does the energy in coal originally come from? Give a detailed explanation. (2)

The energy comes from the sun.✓ Plants long ago absorbed the energy from the Sun for photosynthesis.✓ The remains from the dead plant change into coal.✓

6.4. Fossil fuels are non-renewable resources. Give one positive and one negative to changing to a renewable energy source. (2)

Possible answer

Positive	Less impact on the environment✓
Negative	The cost to switch over to a new system✓

Question 7:

7.1. How many phases of the Moon are there? (1)

Eight phases of the Moon✓

7.2. What is the phase of the Moon called after First Quarter? (1)

Waxing Gibbous✓

7.3. Explain the First Quarter phase of the Moon according to the criteria.

7.3.1. What can be seen? (1)

Half of the Moon is lit; half can be seen. ✓

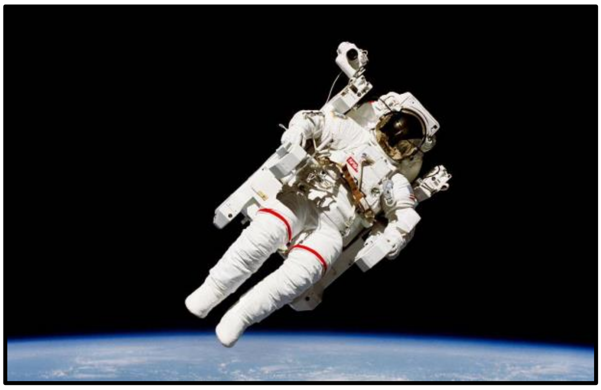
7.3.2. What happens? (1)

During this phase, the Moon is at a right angle to the Sun as seen from Earth. ✓

7.4. What has the most mass in our Solar System? (1)

The Sun✓

7.5. Look at the picture of the astronaut below and answer the questions.



[Astronaut in space, NASA, 27 June 2024, <https://drive.google.com/file/d/1MroBwwAr_uByiIKWfBrCgjsXh-iqa3py/view>]

https://www.nasa.gov/image-article/astronaut-bruce-mccandless-floating-free/

7.5.1. An astronaut in space floats. Does this mean there is no gravity in space? Explain. (2)

There is still gravity in space.✓ However, the astronaut is constantly falling towards Earth along with the spacecraft, creating a feeling of weightlessness.✓

7.5.2. How would the weight of an object on the Moon compare to its weight on Earth? Give a detailed explanation. (2)

The weight of an object on the Moon would be much less than its weight on Earth.✓ The Moon has less mass than Earth, resulting in weaker gravitational pull.✓

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TOTAL: 80